

PatternBoost Three-Problem Final Experiment Analysis

Complete 81-cell trajectories, exact certificate audit, and limitations study

Multi-level PatternBoost project

July 10, 2026

Experimental commit	20750a588a5b5d8e744689b9efa7538b5eaab26b
Slurm array	16602591
Run root	runs/main_81_24h_20260709_1623
Design	3 problems $\times$ 3 representations $\times$ 3 local searches $\times$ 3 surrogates
Repeated-seed axis	None

Abstract

This report gives the final, configuration-level analysis of the 81-cell PatternBoost experiment on three geometric extremal-search problems: the LP integrality gap for maximum independent set of rectangles (MISR), the LP integrality gap for stabbing unit squares by horizontal and vertical lines, and recursive guillotine hardness for disjoint rectangle packings. All cells were run from freshly generated instances under the same 24-hour search budget, with no repeated-seed axis and no warm-started certificate. The best exact scores were 1.5000 for MISR, 1.5000 for unit-square stabbing, and 0.3333 for guillotine hardness. The post-run audit independently re-evaluated the saved certificates: 81 of 81 rows passed and 0 failed. Beyond the endpoint scores, this report analyzes every strict improvement against search generation, elapsed time, and cumulative transformer-training epoch. The main conclusion is that the observed plateaus are primarily structural: representation constraints, discrete objective values, surrogate/exact-score misalignment, and limited compositional mechanisms dominate the effect of additional epochs. By 25% of the model-epoch budget, the final configuration ranking already has descriptive Spearman correlation at least 0.83 in all three problems; median post-best plateaus occupy 74–98% of search generations, and learned-model samples account for only 0–5.4% of strict record improvements.

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Table 1: Component registry for the final matrix.

Problem	Representations	Local search	Surrogates
MISR	endpoint_sequence_pair; triangle_free_rect; quadratic_program_rectangles	sequence_pair_pivot; lp_dual_pivot; program_coeff_pivot	exact_lp_gap_pressure; triangle_free_exact_gap_pressure; graph_conflict_proxy
Unit-square	square_direct; line_square_incidence; sqstab_exact_grid	coord_mutation; primal_dual_lines; sqstab_local_hillclimb	greedy_partial_lp_bitset; exact_stab_gap_pressure; incidence_statistics
Guillotine	rect_direct_disjoint; sequence_pair_packing; recursive_obstruction_grammar	packing_resize; recursive_gadget_assembly; witness_breaking	first_cut_obstruction; depth_limited_dp; k_subset_nonseparability

1 Questions answered by this study

The experiment was designed to answer five distinct questions.

1. What exact value does each of the 81 component combinations reach under the same long-run budget?
2. How does each row improve as search generations and transformer-training epochs accumulate?
3. Which representations, local-search operators, and surrogates are associated with stronger final values, faster discovery, and longer plateaus?
4. Does the learned model contribute new record candidates, or does most progress come from initialization and local mutation?
5. Why did the search fail to move beyond the simple observed barriers, and which limitations are computational rather than mathematical?

The report deliberately separates three kinds of statements. An **exact result** is recomputed from a standalone certificate. A **descriptive matrix comparison** compares the 27 different configurations for one problem. A **mechanistic interpretation** uses both the trajectories and the implementation to explain a pattern. Only the first category is a certificate-level mathematical claim.

2 Experimental design and evidence standard

2.1 The full factorial matrix

For each problem, the matrix crosses three representations, three local-search operators, and three surrogate objectives. Thus each problem has $3^3 = 27$ cells and the complete experiment has 81 cells. This is a full component factorial, not 27 repetitions of one algorithm.

2.2 What every component means

Across the complete study there are nine representation families, nine local-search operators, and nine surrogate objectives. Each problem uses its own three of each; only those three are crossed, giving $3 \times 3 \times 3 = 27$ valid cells per problem. A representation specifies emphhow an instance is encoded

and repaired; local search specifies *emph* how an encoded instance is changed; and a surrogate specifies *emph* how candidates are ranked before expensive exact verification. The exact verifier, not a surrogate, decides whether a certificate becomes a saved record.

Table 2: Representation glossary.

Problem	Representation	Meaning
MISR	<code>endpoint_sequence_pair</code>	Two double-occurrence sequences encode the endpoint orders on the horizontal and vertical axes. A deterministic decoder turns the pair into rectangles, so mutations act on relative endpoint order rather than raw coordinates.
MISR	<code>triangle_free_rect</code>	Direct rectangle coordinates, with generation and repair restricted so that the rectangle intersection graph is triangle-free. It searches sparse conflict graphs directly.
MISR	<code>quadratic_program_rectangles</code>	A bounded integer coefficient program, decoder choice, and block composition are decoded to rectangles and then repaired to the permitted size and triangle-free constraint.
Unit-square	<code>square_direct</code>	Direct lower-left grid coordinates for unit squares (or the stored common side length). This is the least structured geometric encoding.
Unit-square	<code>line_square_incidence</code>	A balanced row/column incidence layout is generated first and then mapped to square coordinates. The stored payload tracks which abstract grid row and column each square occupies.
Unit-square	<code>sqstab_exact_grid</code>	Distinct finite-grid square coordinates plus a common side length, normalized to a canonical origin after repair. It is designed to make the exact finite line-stabbing geometry easy to audit.
Guillotine	<code>rect_direct_disjoint</code>	Direct coordinates for disjoint rectangles, followed by geometric repair that removes overlaps and enforces the size bounds.
Guillotine	<code>sequence_pair_packing</code>	A shuffled order drives a shelf-packing decoder that places small disjoint rectangles row by row. The order supplies the combinatorial packing prior.
Guillotine	<code>recursive_obstruction_grammar</code>	Randomly instantiated non-sliceable obstruction cores are combined with filler rectangles and then repaired. It is a random generator family, not a fixed injected certificate.

Table 3: Local-search glossary.

Problem	Local search	Meaning
MISR	<code>sequence_pair_pivot</code>	Swaps or reorders endpoints in the sequence-pair encoding, including occasional size lifts. Candidate neighbors are compared with a MISR LP-gap/independent-set key before repair.

Problem	Local search	Meaning
MISR	<code>lp_dual_pivot</code>	Uses the same representation-safe mutation families but emphasizes neighbors favored by the exact LP-gap and independent set signals; in direct triangle-free coordinates it preferentially adds or replaces rectangles that preserve sparse conflicts.
MISR	<code>program_coeff_pivot</code>	When a program payload is present, perturbs coefficient blocks, decoder choice, composition, or active size. In other representations it falls back to their safe local mutation templates.
Unit-square	<code>coord_mutation</code>	Basic coordinate-level move, duplicate, or deletion of a square, followed by canonical geometric repair.
Unit-square	<code>primal_dual_lines</code>	Biases moves toward aligning a square with another square’s horizontal or vertical coordinate, so a proposed change directly affects candidate stabbing lines.
Unit-square	<code>sqstab_local_hillclimb</code>	Builds several nearby line-tightening, addition/replacement, deletion, restart, and canonical-shift candidates, then retains the strongest quick local neighbor.
Guillotine	<code>packing_resize</code>	Moves, resizes, adds, replaces, and occasionally bars rectangles, then applies a nonseparability-guided local step. It is a geometry-first packing search.
Guillotine	<code>recursive_gadget_assembly</code>	Inserts or copies small obstruction motifs, bars, and recursive tiles to test whether local hard patterns can be assembled into a larger packing.
Guillotine	<code>witness_breaking</code>	Uses a separable-subset/cut witness to close a gap or add a bridge/blocker that makes the witness’s easy guillotine cut less effective.

Table 4: Surrogate-objective glossary.

Problem	Surrogate	Meaning
MISR	<code>exact_lp_gap_pressure</code>	Calls the MISR exact scorer during ranking and uses the LP/integral gap, with a small triangle-free lower-bound tie-break when applicable. It is the most aligned, most expensive MISR ranking signal.
MISR	<code>triangle_free_exact_gap_pressure</code>	Rejects non-triangle-free candidates, then ranks the remaining candidates by their exact MISR gap with small density and certified-lower-bound tie-breaks.
MISR	<code>graph_conflict_proxy</code>	A cheap graph statistic based on intersection-edge density, a maximum-independent-set hint, maximum clique size, and a triangle-free bonus.
Unit-square	<code>greedy_partial_lp_bitset</code>	Approximates the gap by dividing a greedy stabbing cover size by a max-line-frequency fractional lower bound computed from incidence bitsets.
Unit-square	<code>exact_stab_gap_pressure</code>	Calls the exact stabbing solver during ranking and combines the exact gap with small terms favoring a large integral cover, low LP value, and nontrivial LP support.

Problem	Surrogate	Meaning
Unit-square	<code>incidence_statistics</code>	Uses greedy cover size, the number of candidate stabbing lines, and balance of line frequencies; it is a cheap incidence-only proxy.
Guillotine	<code>first_cut_obstruction</code>	Measures how many rectangles the best first axis-parallel cut must cross, with bonuses for balanced partitions and comparable horizontal/vertical projection spans.
Guillotine	<code>depth_limited_dp</code>	Runs a depth-four dynamic-programming estimate of how many rectangles a shallow guillotine strategy can save, and ranks by its implied destroyed fraction.
Guillotine	<code>k_subset_nonseparability</code>	Tests whether majority-sized subsets resist guillotine separation (exactly for small n , sampled otherwise) and combines that pressure with shallow-DP, beam, balance, projection, and size terms.

Each cell used an independently generated random seed recorded in the matrix manifest. Seeds are provenance fields, not an experimental axis: there is one run per component cell. Consequently, marginal means and rank correlations in this report are descriptive summaries of the matrix. They are not confidence intervals over repeated stochastic trials.

2.3 PatternBoost loop

Every cell repeatedly performs the following cycle.

1. Generate or repair a population in the cell’s representation.
2. Score every novel candidate with the chosen surrogate.
3. Exactly audit the elite candidates at the configured interval.
4. Save a certificate only when the exact score strictly improves.
5. Mutate elites with the chosen local operator.
6. Every ten search generations, initialize and train a new small transformer for three epochs on the strongest valid archive entries and sample additional candidates from it.

The model never sees a target answer. Its training corpus is the serialized archive selected by the current surrogate. Exact verification remains outside the model and controls certificate acceptance. Model parameters are not carried from one training event to the next; each event is a fresh three-epoch fit. When transformer sampling does not yield enough parseable candidates, the implementation may fill the shortfall with an archive-trained character n-gram sampler.

2.4 What “epoch” means in the plots

Three progress axes are retained because they answer different questions.

- **Search generation** counts complete population updates. It is the most direct algorithmic progress variable, but expensive scorers complete fewer generations per hour.

- **Elapsed time** is the fairest resource comparison because every cell received the same wall-clock budget.
- **Cumulative transformer-training epoch** counts three epochs for every successful model-training event. Because each event trains a newly initialized model, this is an accounting axis over repeated short fits, not a standard learning curve for one continuously trained network. It should not be confused with search generation. A score can improve between training events through local mutation.

All three views are plotted. The model-epoch plot therefore measures where improvements occur relative to learned-model updates, while the generation and time plots guard against misleading conclusions caused by different scorer speeds.

2.5 Nontriviality and natural-search constraints

The final code enforced at least eight geometric objects in every accepted training example. Unit-square certificates additionally required integer stabbing number at least four, and guillotine certificates required at least two destroyed rectangles. The run did not start from any previously saved best certificate. All 81 checkpoints contained a naturally discovered best certificate before finalization; the guillotine emergency export fallback was not used.

The search is not entirely prior-free. Endpoint orders, balanced incidence layouts, triangle-free repair, and recursive obstruction layouts encode domain knowledge. In particular, the guillotine `recursive_obstruction_grammar` contains a randomized geometric realization of a fixed seven-rectangle spiral topology. It does not encode a score or final certificate, but it is a strong construction-family prior and must be disclosed when interpreting representation effects.

3 Run integrity and exact audit

The source deployed to Jubail matched commit `20750a588a5b5d8e744689b9efa7538b5eaab26b`. The matrix manifest, event streams, checkpoints, summaries, certificates, renderings, Slurm logs, compact trajectory tables, and analysis-script hashes are retained under one run root. The strict post-processing job checked:

1. exactly 81 unique component cells and 27 cells per problem;
2. one completed summary and checkpoint per cell;
3. clean stop reason, return code, and elapsed-time fields;
4. a nonempty best-certificate path for every cell;
5. agreement between event-stream, checkpoint, summary, and certificate scores;
6. certificate hash consistency; and
7. independent exact verification of every best certificate.

All 81 array tasks completed with return code zero, stop reason `budget_exhausted`, and empty stderr. Observed row elapsed times ranged from 86,405 to 88,007 seconds. The longest row exceeded the

Table 5: Overall outcome of the completed 81-cell matrix. Quantiles are across the 27 component configurations, not repeated random seeds.

Problem	Min.	Q1	Median	Q3	Best	Cells at best
MISR	$\frac{9}{8}$	$\frac{5}{4}$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{3}{2}$	1
Unit-square	$\frac{6}{5}$	1.2829	$\frac{35}{26}$	$\frac{10}{7}$	$\frac{3}{2}$	4
Guillotine	$\frac{2}{13}$	0.2426	$\frac{1}{4}$	$\frac{3}{11}$	$\frac{1}{3}$	1

nominal 86,400-second search budget by roughly 26 minutes because it entered a hard MISR solve before the next between-generation budget check; it still finished inside the 24-hour-30-minute Slurm allocation and wrote a normal summary.

The audit outcome was 81 passed and 0 failed out of 81 rows. A row that fails any of these checks is retained in the diagnostic appendix but is not evidence for a reported construction.

4 Overall results

The best values alone do not describe the experiment. MISR has median 1.3333 and range 1.1250–1.5000; unit-square stabbing has median 1.3462 and range 1.2000–1.5000; guillotine hardness has median 0.2500 and range 0.1538–0.3333. The number of cells attaining the best value is 1, 4, and 1, respectively. These differences distinguish a robust plateau from a rare configuration-specific hit.

Table 6: Top five configurations per problem. Epoch is cumulative transformer training epoch; time is elapsed hours to the first appearance of the final best.

Problem	Score	Representation	Local search	Surrogate	Epoch	Time	Items
MISR	$\frac{3}{2}$	triangle_free_rect	sequence_pair_pivot	exact_lp_gap_pressure	1860	3.12	12
MISR	$\frac{7}{5}$	triangle_free_rect	program_coeff_pivot	exact_lp_gap_pressure	4782	14.48	14
MISR	$\frac{11}{8}$	triangle_free_rect	lp_dual_pivot	triangle_free_exact_gap_pressure	24	0.03	11
MISR	$\frac{11}{8}$	triangle_free_rect	lp_dual_pivot	exact_lp_gap_pressure	81	0.08	11
MISR	$\frac{11}{8}$	triangle_free_rect	program_coeff_pivot	triangle_free_exact_gap_pressure	156	0.15	11
Unit-square	$\frac{3}{2}$	sqstab_exact_grid	coord_mutation	exact_stab_gap_pressure	117	0.35	14
Unit-square	$\frac{3}{2}$	line_square_incidence	sqstab_local_hillclimb	exact_stab_gap_pressure	3003	9.29	21
Unit-square	$\frac{3}{2}$	line_square_incidence	coord_mutation	exact_stab_gap_pressure	3915	11.18	21
Unit-square	$\frac{3}{2}$	sqstab_exact_grid	sqstab_local_hillclimb	exact_stab_gap_pressure	375	0.46	19
Unit-square	$\frac{45}{31}$	square_direct	sqstab_local_hillclimb	exact_stab_gap_pressure	2124	3.68	24
Guillotine	$\frac{1}{3}$	rect_direct_disjoint	witness_breaking	depth_limited_dp	858	5.81	9
Guillotine	$\frac{3}{10}$	sequence_pair_packing	witness_breaking	depth_limited_dp	6	0.05	10

Problem	Score	Representation	Local search	Surrogate	Epoch	Time	Items
Guillotine	$\frac{3}{10}$	sequence_pair_ packing	witness_breaking	first_cut_ obstruction	669	3.51	10
Guillotine	$\frac{3}{11}$	rect_direct_ disjoint	packing_resize	depth_limited_dp	345	1.92	11
Guillotine	$\frac{3}{11}$	sequence_pair_ packing	witness_breaking	k_subset_ nonseparability	45	2.12	22

Final exact score for every matrix cell

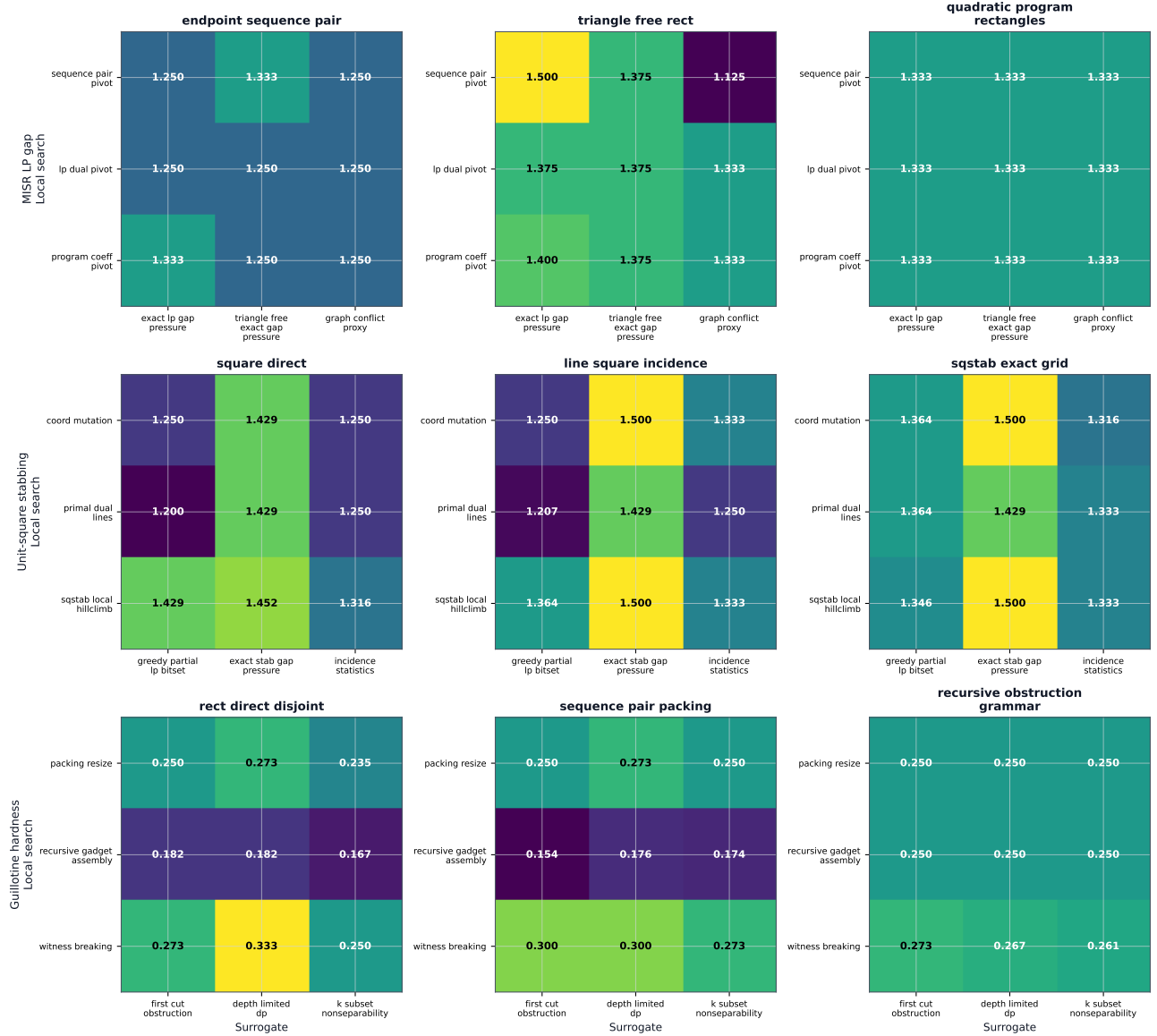


Figure 1: Final exact score in every cell. Rows within each panel are local searches, columns are surrogates, and the three panels in each problem row are representations. Color ranges are normalized within a problem; printed numbers are the evidence.

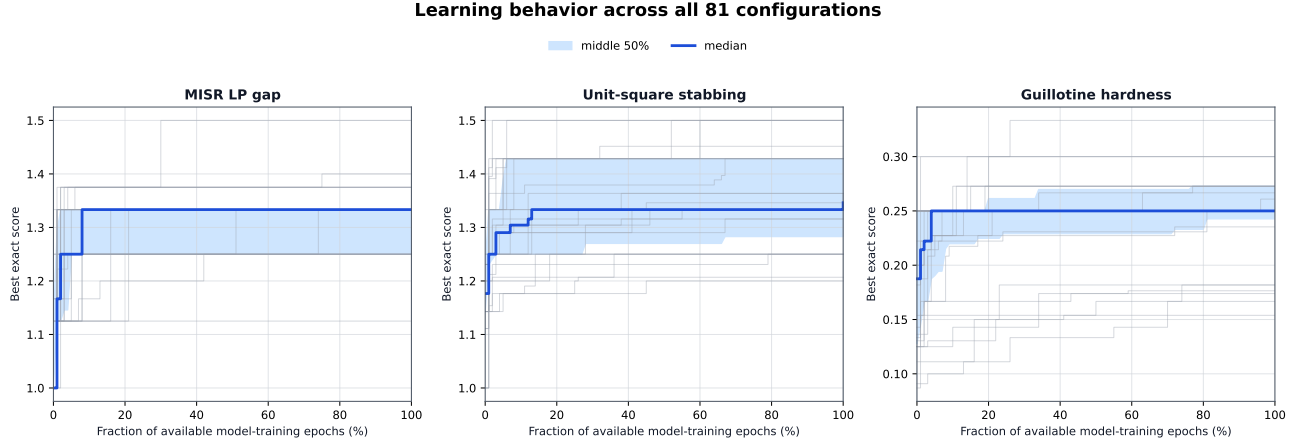


Figure 2: All 27 trajectories per problem after normalizing each cell by its available cumulative transformer-training epochs. Gray lines are individual configurations; the blue line is the median and the band is the middle 50%. This normalization compares the shape of learning, not raw computational throughput.

5 Learning dynamics across all 81 cells

5.1 Normalized model-epoch behavior

A long horizontal segment means that additional model updates and local search failed to improve the exact record. A late jump can mean either that the model found a new structural combination or that local mutation crossed a rare discrete objective threshold. The source analysis below separates those cases.

5.2 Early prediction of the final ranking

At 25% of the epoch budget, the descriptive rank correlations with the final ranking are 0.834 for MISR, 0.916 for unit-square stabbing, and 0.941 for guillotine hardness. High early correlation indicates that longer runtime mostly refines an ordering already visible early; low correlation indicates meaningful late rearrangement among configurations.

All three correlations exceed 0.83 by this point. Unit-square and guillotine rankings are already strongly predictive after only 10% of the epoch budget, while MISR reorganizes somewhat longer. The remaining runtime mostly extends plateaus or improves a small number of outlying cells rather than changing the broad component ordering.

5.3 Where strict improvements came from

The learned-model path contributes to discovery only when it generates a valid candidate that survives representation repair, satisfies nontriviality, and beats the exact incumbent. The event schema labels these candidates `model_sample`; it does not distinguish a direct transformer sample from an n-gram backfill. The corresponding fractions of all strict improvements are 0.0%, 1.2%, and 5.4%. This statistic is more informative than the raw number of training calls: a model can train thousands of times while mostly reproducing the archive or proposing invalid candidates. In absolute terms, the event streams contain no model-sourced strict MISR improvement, only two for unit-square stabbing, and

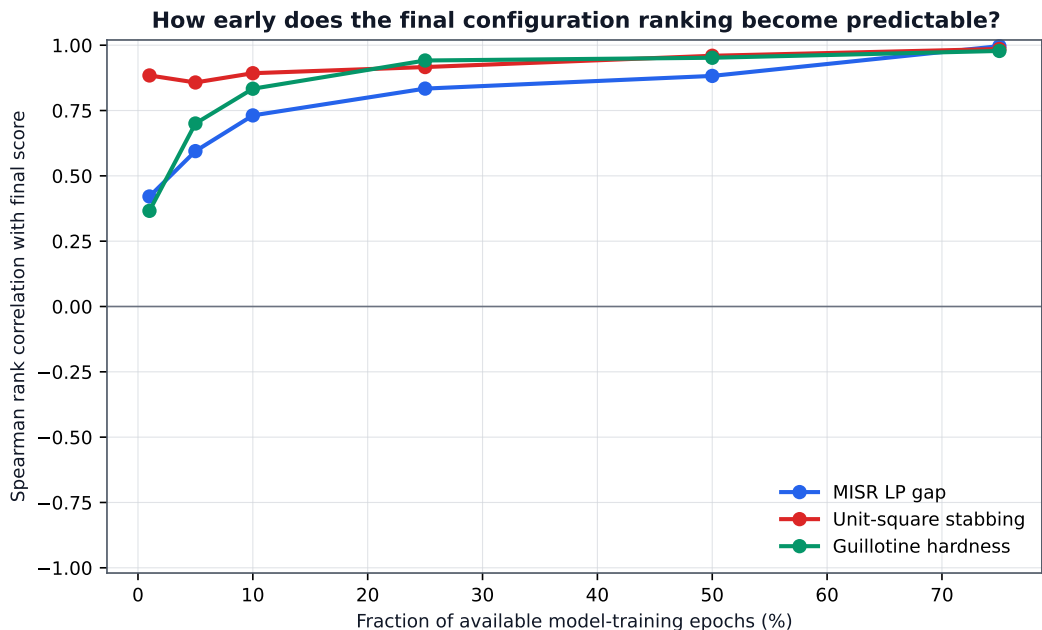


Figure 3: Spearman rank correlation between the configuration ranking at an early fraction of the model-epoch budget and the final ranking. Because each cell is a different configuration and there are no repeated seeds, the curve is descriptive rather than an inferential test.

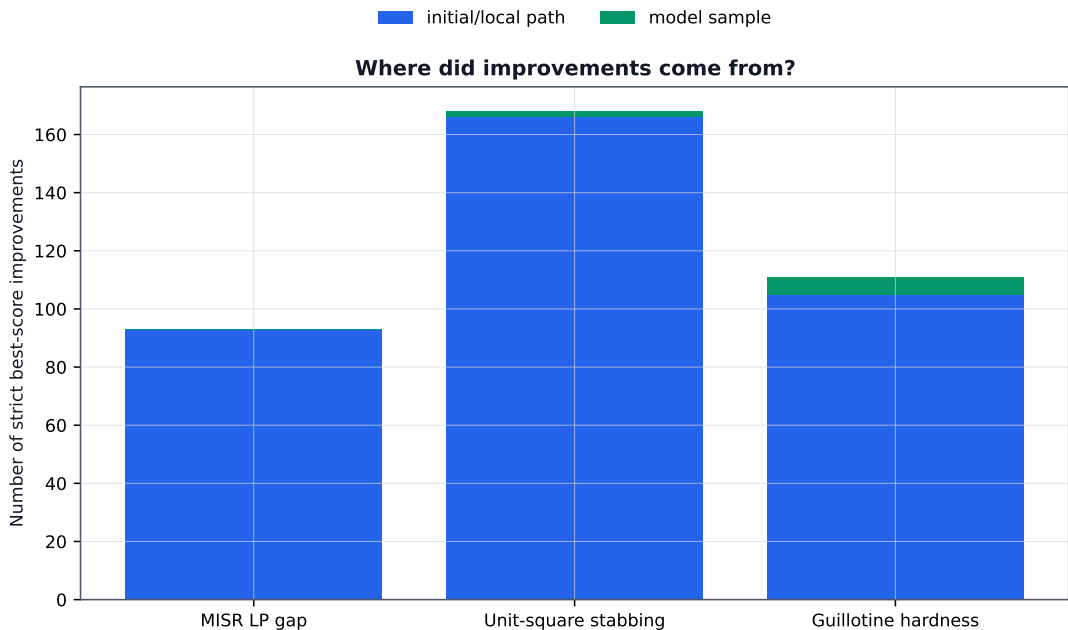


Figure 4: Source of every strict best-score improvement recorded in the 81 event streams. A model sample is counted only when that sampled candidate is exactly audited and strictly improves the saved record.

six for guillotine. The combined initialization/local-mutation path accounts for 93, 166, and 105 strict improvements, respectively. The combined label is deliberate: `codeinitial_or_local` events do not support a finer attribution.

5.4 Plateaus and computational throughput

The median post-best plateau occupies 97.5% of MISR generations, 74.4% of unit-square generations, and 80.5% of guillotine generations. The top records first appeared after 3.12, 0.35, and 5.81 hours.

The median cell produced 17.16 duplicate candidates per MISR generation, 9.46 per unit-square generation, and 7.25 per guillotine generation, from a population of 32. Median valid learned-model yields were only 39.2% and 38.8% for the two LP-gap problems, versus 94.9% for guillotine. High syntactic validity therefore does not imply useful novelty: guillotine proposals were often repairable, yet their exact scores still plateaued.

Plateau diagnostics

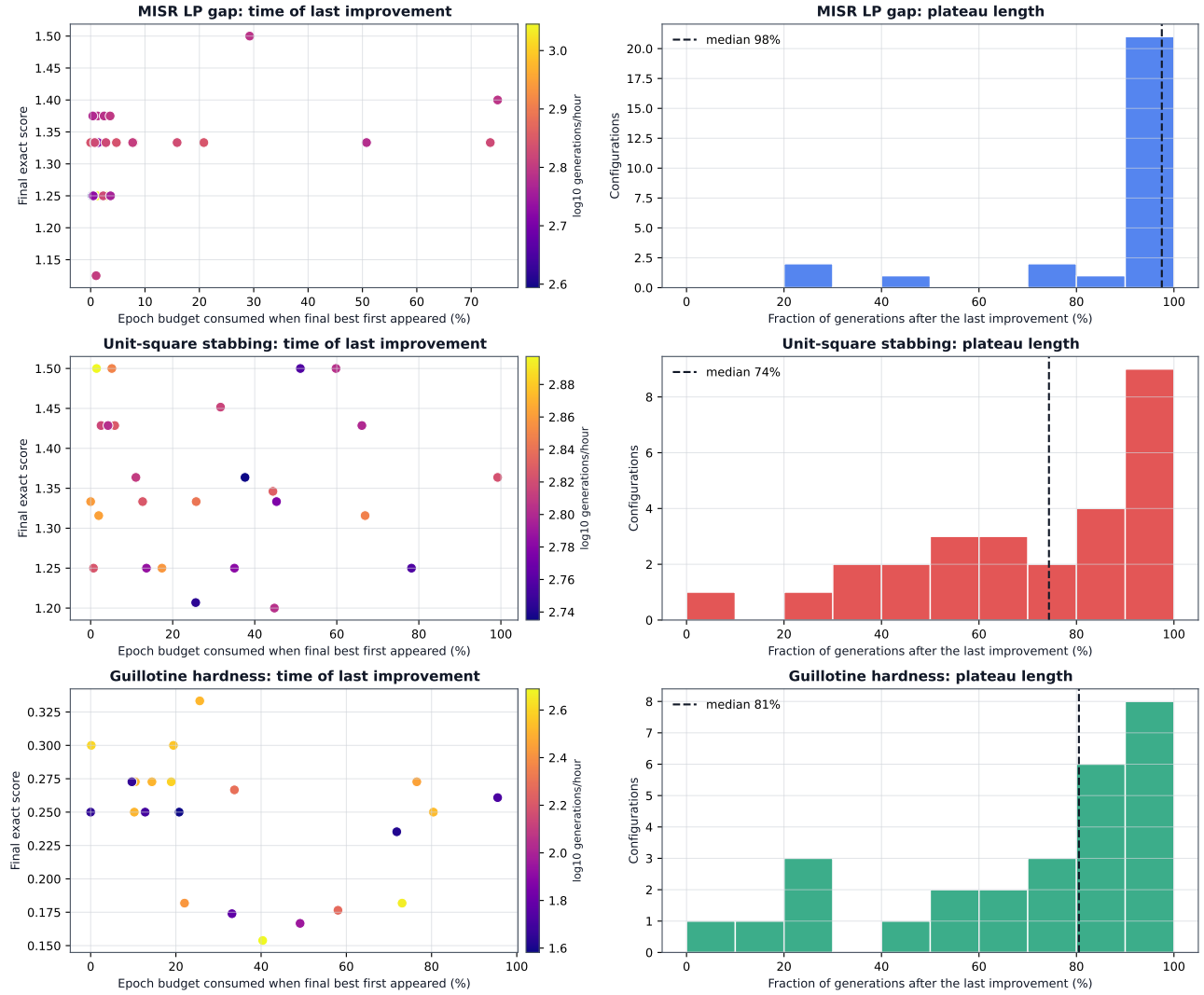


Figure 5: Left: when each cell first reached its final score, colored by search throughput. Right: fraction of search generations after the final strict improvement. Long plateaus provide direct evidence that wall time alone was not the principal missing ingredient.

Search throughput versus solution quality

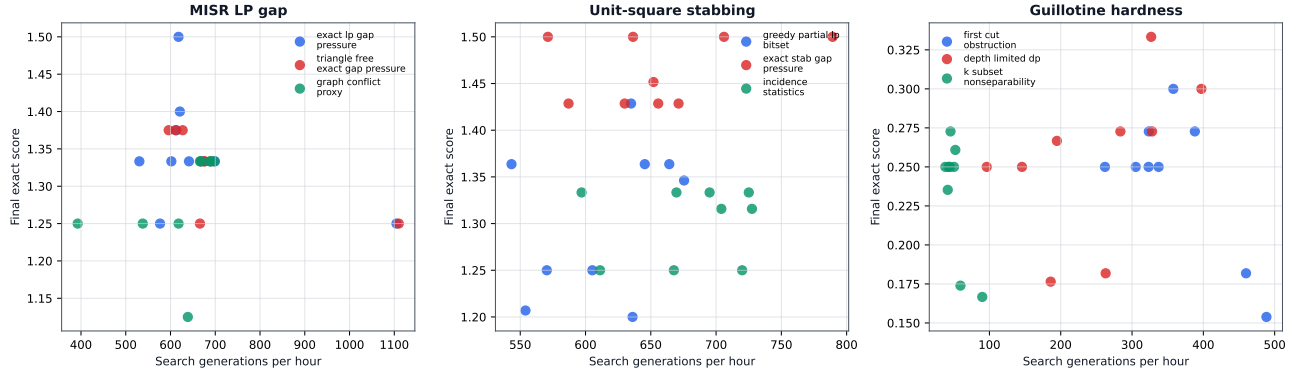


Figure 6: Search generations completed per hour versus final exact score. Exact or deeper surrogates may lower throughput while improving alignment; fast proxies are useful only if their additional generations translate into better exact certificates.

Search-efficiency diagnostics: duplication, model yield, and repair pressure

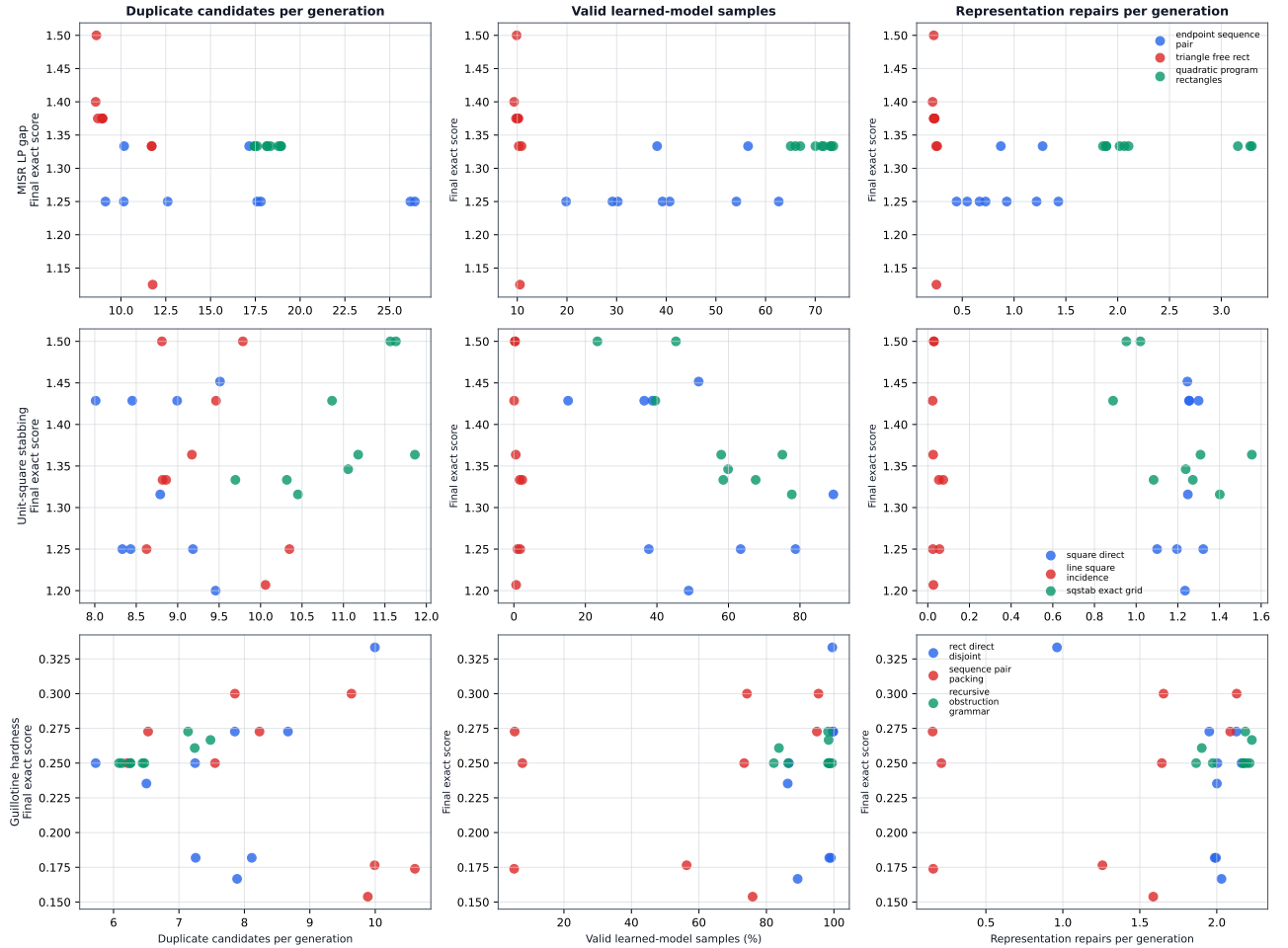


Figure 7: Final score against duplicate pressure, valid learned-model sample yield, and representation-repair pressure. These diagnostics test whether additional training produces genuinely new feasible structures or repeatedly returns to the same basin.

Table 7: Descriptive component marginals for MISR. Each level pools nine distinct component cells.

Factor	Level	Mean	Median	Min.	Max.	IQR
Representation	<code>triangle_free_rect</code>	1.3546	1.3750	1.1250	1.5000	[1.3333, 1.3750]
	<code>quadratic_program_rectangles</code>	1.3333	1.3333	1.3333	1.3333	[1.3333, 1.3333]
	<code>endpoint_sequence_pair</code>	1.2685	1.2500	1.2500	1.3333	[1.2500, 1.2500]
Local Search	<code>program_coeff_pivot</code>	1.3269	1.3333	1.2500	1.4000	[1.3333, 1.3333]
	<code>lp_dual_pivot</code>	1.3148	1.3333	1.2500	1.3750	[1.2500, 1.3333]
	<code>sequence_pair_pivot</code>	1.3148	1.3333	1.1250	1.5000	[1.2500, 1.3333]
Surrogate	<code>exact_lp_gap_pressure</code>	1.3454	1.3333	1.2500	1.5000	[1.3333, 1.3750]
	<code>triangle_free_exact_gap_pressure</code>	1.3287	1.3333	1.2500	1.3750	[1.3333, 1.3750]
	<code>graph_conflict_proxy</code>	1.2824	1.3333	1.1250	1.3333	[1.2500, 1.3333]

6 Problem 1: MISR LP-gap search

6.1 Objective and exact verifier

For a finite family $\mathcal{R}$ of axis-parallel rectangles, the exact score is

$$\text{score}_{\text{MISR}}(\mathcal{R}) = \frac{\alpha^*(\mathcal{R})}{\alpha(\mathcal{R})},$$

where α is the maximum number of pairwise-disjoint rectangles and α^* is the clique-LP optimum of the rectangle intersection graph. The certificate stores all rectangles, the intersection graph, maximal cliques, an optimal integral independent set, LP primal and dual values, and solver status. The verifier reconstructs and solves both optimization problems.

6.2 All 27 trajectories

The matrix separates the effects cleanly at a descriptive level. The `triangle_free_rect` representation has mean score 1.3546 and is the only representation to reach 1.5. Every `quadratic_program_rectangles` cell ends at exactly 4/3, a striking sign that its decoder and triangle-free repair map many different component choices into the same objective basin. The unrestricted endpoint-sequence representation averages only 1.2685. Surrogate alignment is also visible: `exact_lp_gap_pressure` averages 1.3454, compared with 1.2824 for `graph_conflict_proxy`. By contrast, the three local-search marginal means differ by only about 0.012, so representation and scoring dominate the choice of mutation label in this run.

6.3 What worked

The strongest MISR behavior combines a representation that can maintain sparse conflict structure with a surrogate that observes the exact LP/integer imbalance. Triangle-free repair removes three-way cliques, making edge constraints the clique LP and exposing half-integral pressure. Endpoint sequence pairs provide large combinatorial moves without tying the search to small coordinate perturbations.

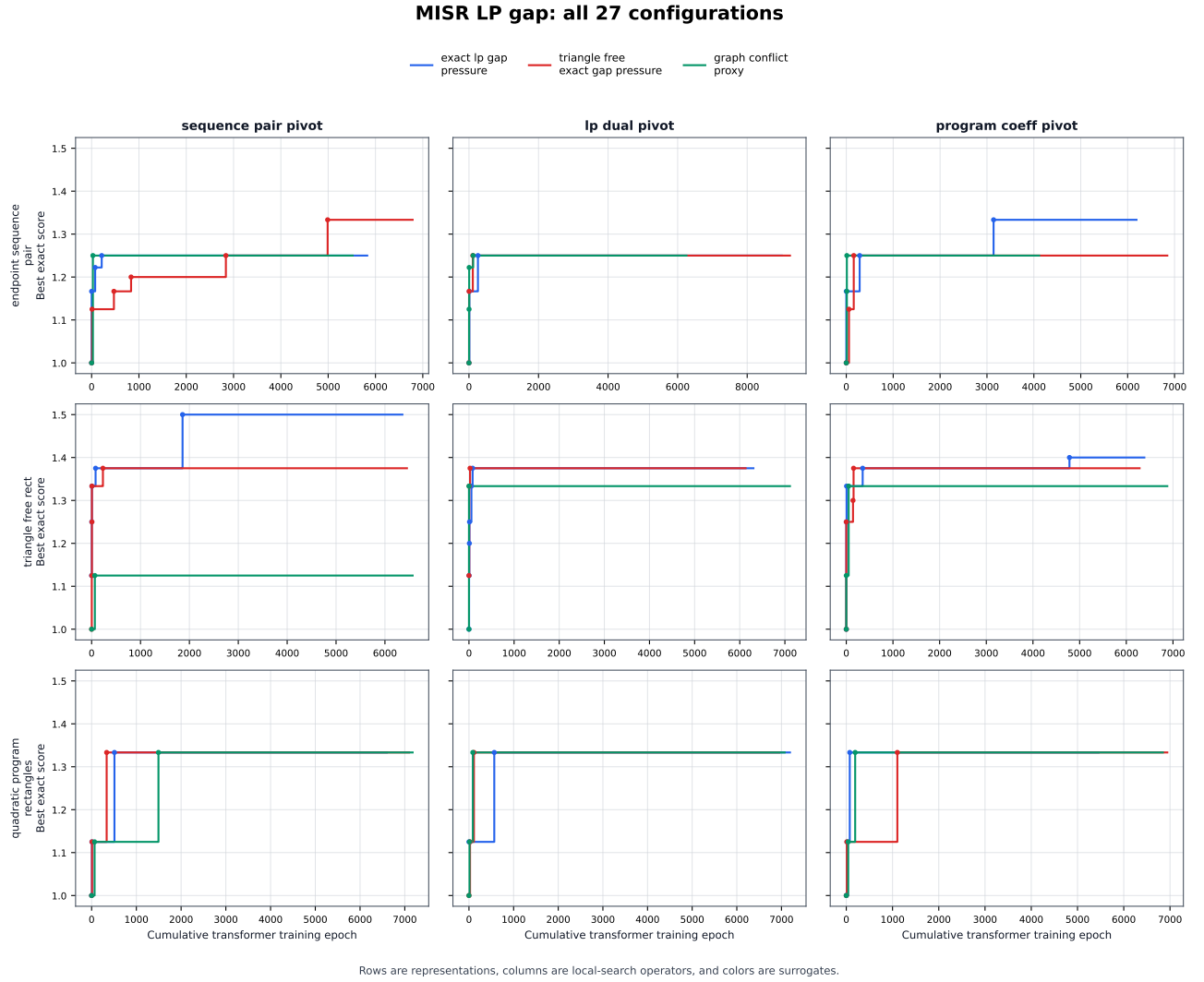


Figure 8: MISR trajectories against cumulative transformer-training epoch. Each panel fixes a representation and local search; its three colored curves are the surrogates.

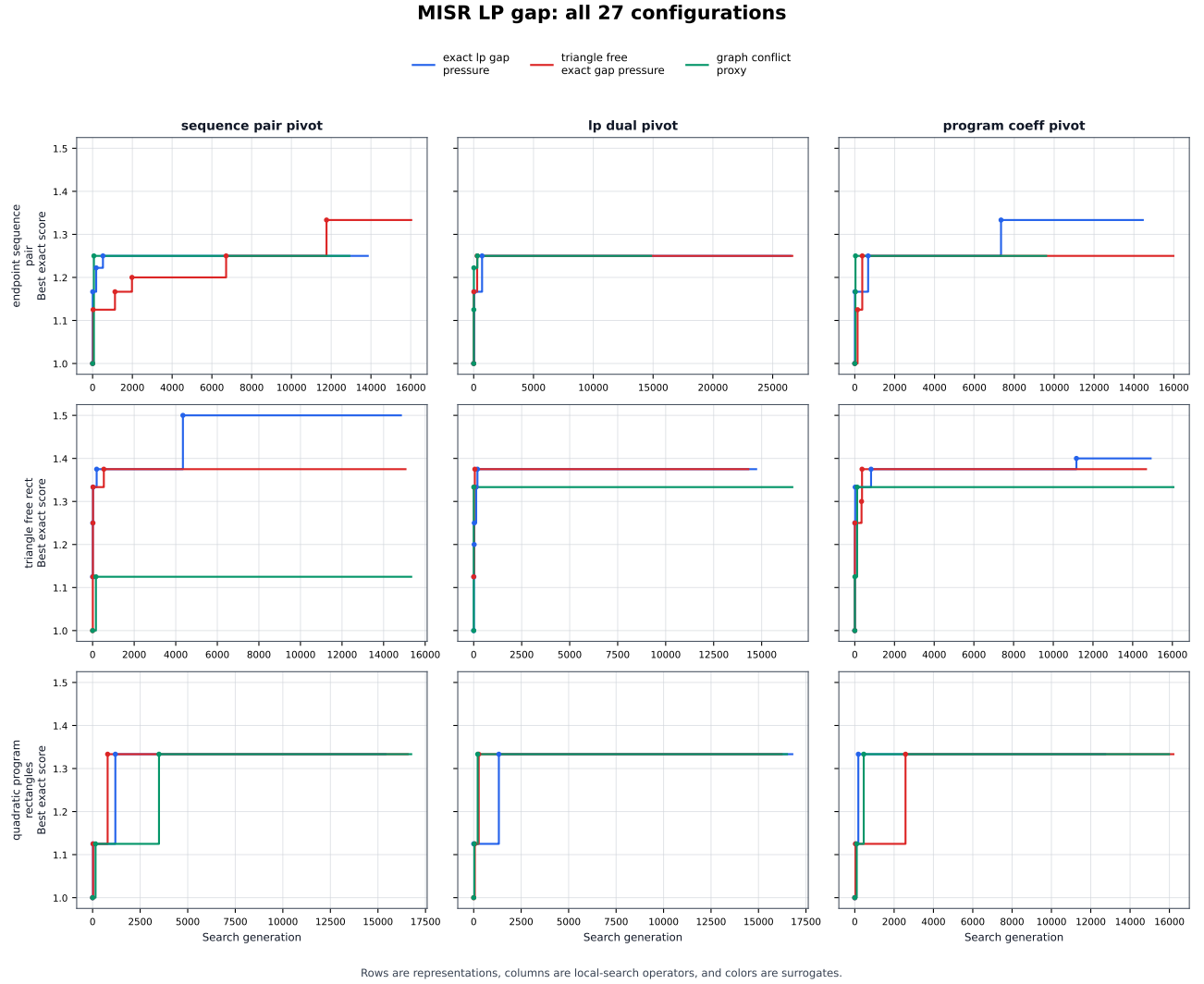


Figure 9: The same 27 MISR cells against search generation. Comparing this figure with Figure 8 separates model-update effects from raw local-search progress.

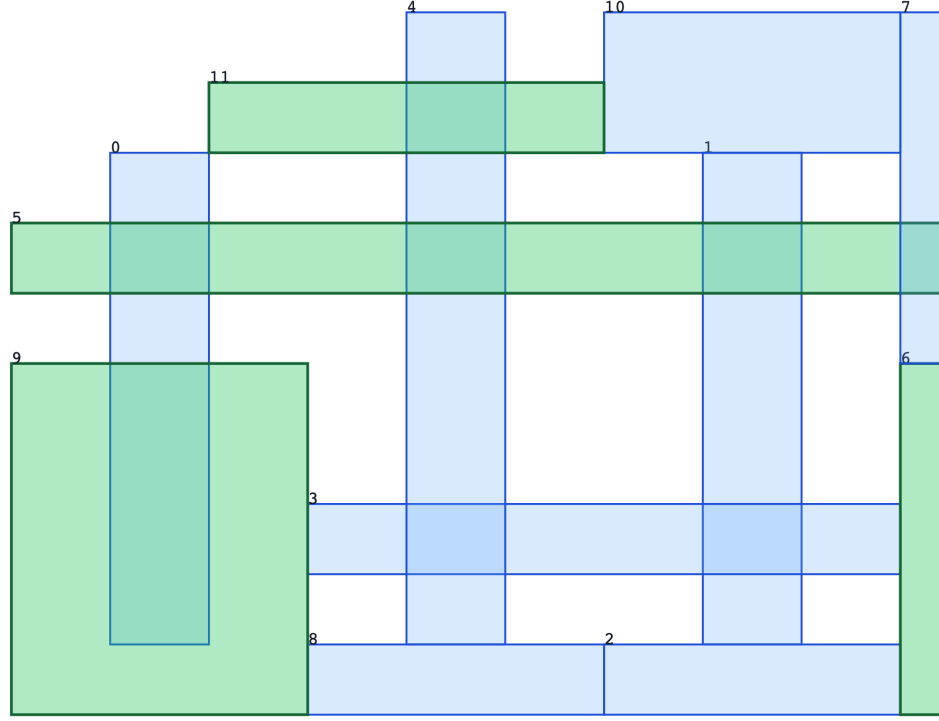


Figure 10: Best audited MISR construction from the final matrix. The drawing is explanatory; the JSON certificate and exact verifier are the evidence.

LP-aware pivots are useful when they change tight constraints without simultaneously increasing the integral optimum.

6.4 Why improvement stopped

Four mechanisms explain the MISR plateau.

1. **Two objectives must move in opposite directions.** Adding intersections can raise LP pressure but often also increases the maximum independent-set structure in a way that leaves the ratio unchanged.
2. **Triangle-free search has a built-in ceiling mechanism.** The clique LP becomes the edge relaxation, and many repaired graphs collapse to common half-integral ratios. Repair can delete exactly the vertices that would have enabled a more unusual denominator.

3. **The representations lack a reliable composition operator.** Duplicating a good finite motif usually scales α and α^* at the same rate, preserving rather than improving the ratio.
4. **Model training is surrogate-conditioned.** If the elite archive contains many geometrically different encodings of the same conflict graph, the transformer learns syntax and local motif frequency more easily than the rare global condition that improves α^*/α .

The result is a discovery method for compact finite witnesses, not yet an amplification mechanism for a new asymptotic lower bound.

Table 8: Descriptive component marginals for Unit-square. Each level pools nine distinct component cells.

Factor	Level	Mean	Median	Min.	Max.	IQR
Representation	<code>sqstab_exact_grid</code>	1.3872	1.3636	1.3158	1.5000	[1.3333, 1.4286]
	<code>line_square_incidence</code>	1.3518	1.3333	1.2069	1.5000	[1.2500, 1.4286]
	<code>square_direct</code>	1.3337	1.3158	1.2000	1.4516	[1.2500, 1.4286]
Local Search	<code>sqstab_local_hillclimb</code>	1.3969	1.3636	1.3158	1.5000	[1.3333, 1.4516]
	<code>coord_mutation</code>	1.3546	1.3333	1.2500	1.5000	[1.2500, 1.4286]
	<code>primal_dual_lines</code>	1.3211	1.3333	1.2000	1.4286	[1.2500, 1.4286]
Surrogate	<code>exact_stab_gap_pressure</code>	1.4629	1.4516	1.4286	1.5000	[1.4286, 1.5000]
	<code>greedy_partial_lp_bitset</code>	1.3081	1.3462	1.2000	1.4286	[1.2500, 1.3636]
	<code>incidence_statistics</code>	1.3017	1.3158	1.2500	1.3333	[1.2500, 1.3333]

7 Problem 2: unit-square line stabbing

7.1 Objective and exact verifier

For a family $\mathcal{S}$ of congruent axis-parallel squares,

$$\text{score}_{\text{stab}}(\mathcal{S}) = \frac{\tau(\mathcal{S})}{\tau^*(\mathcal{S})},$$

where τ is the minimum number of horizontal and vertical lines stabbing all squares and τ^* is the LP optimum over the compressed critical-line universe. The certificate stores square coordinates, side length, critical lines, line-square incidence, an optimal integer cover, LP primal and dual solutions, and solver status.

7.2 All 27 trajectories

Unit-square stabbing has the strongest single component effect in the entire study. The exact gap surrogate averages 1.4629 and every one of its nine cells lies between $10/7$ and $3/2$. The greedy and incidence-statistics surrogates average only 1.3081 and 1.3017. Thus additional generations under a cheap proxy do not compensate for optimizing the wrong quantity. The `sqstab_exact_grid` representation has the strongest representation mean (1.3872), while `sqstab_local_hillclimb` has the strongest local search mean (1.3969). Four exact-pressure cells reach numerical $3/2$, showing that this plateau is reproducible across multiple component paths even without repeated seeds.

7.3 What worked

The successful paths optimize the geometric set-cover instance through its critical-line incidence rather than treating coordinates as independent numbers. Exact gap pressure is particularly valuable because greedy cover size, maximum line frequency, and incidence balance are not reliable proxies for the LP

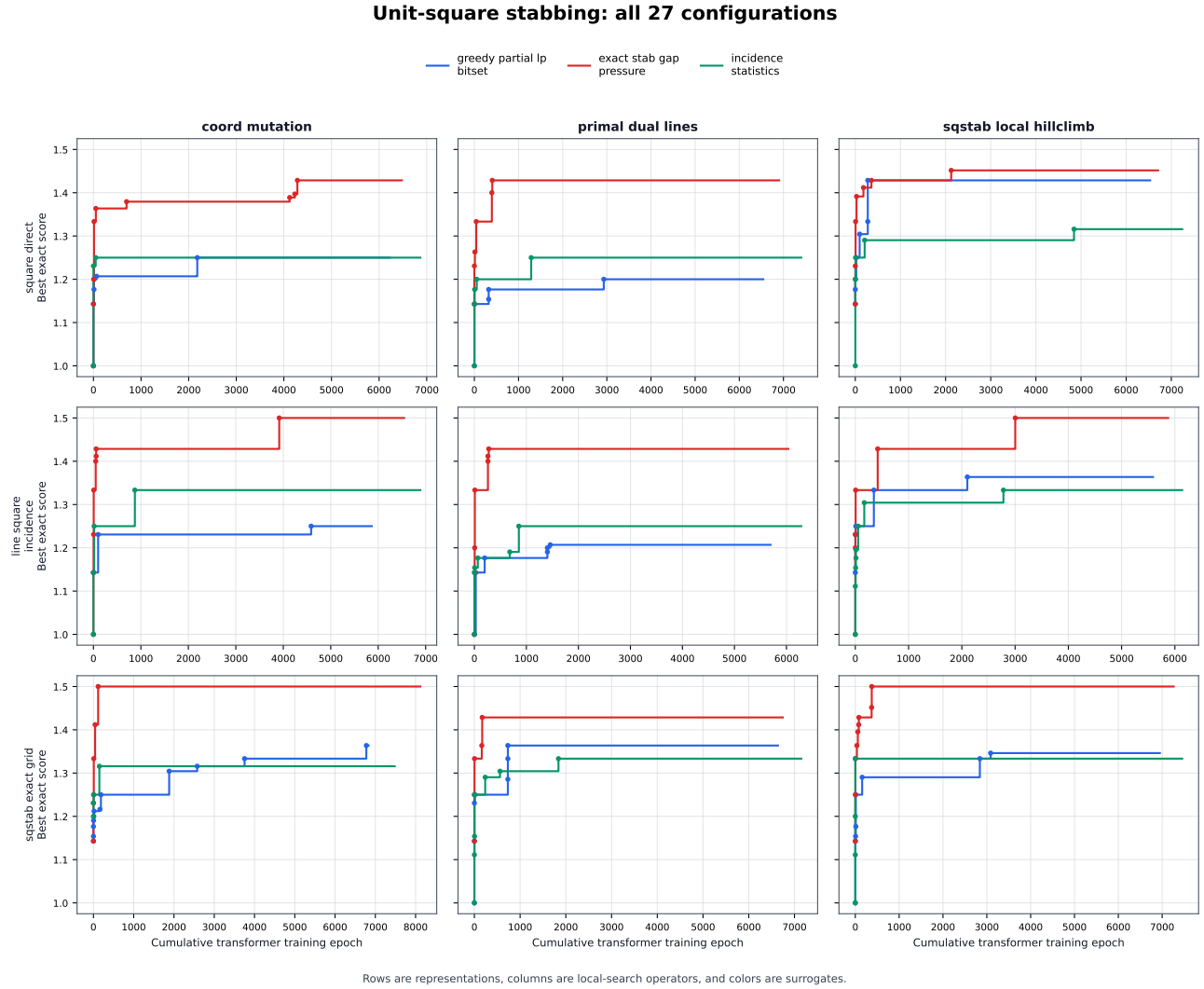


Figure 11: Unit-square trajectories against cumulative transformer-training epoch.

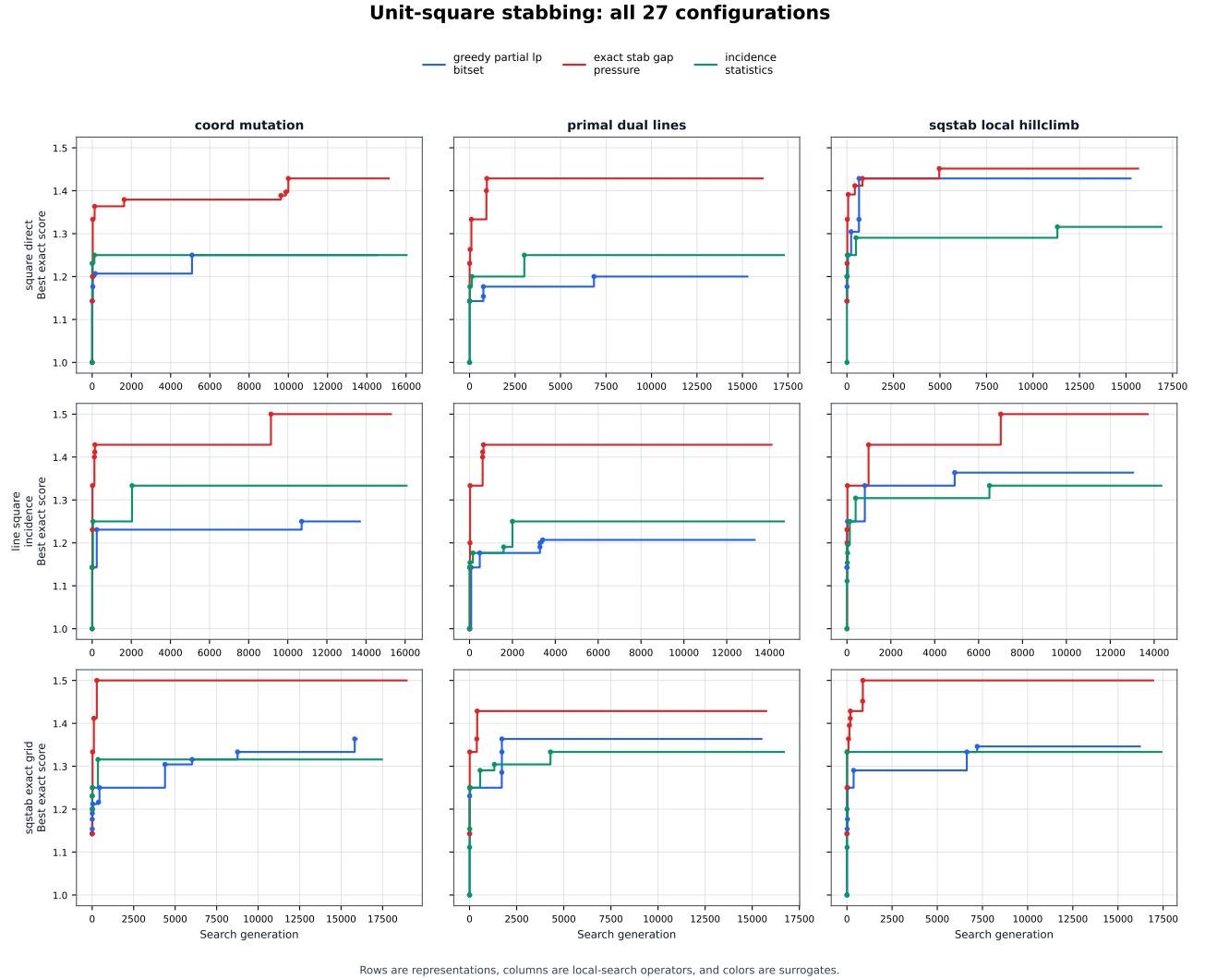


Figure 12: Unit-square trajectories against search generation.

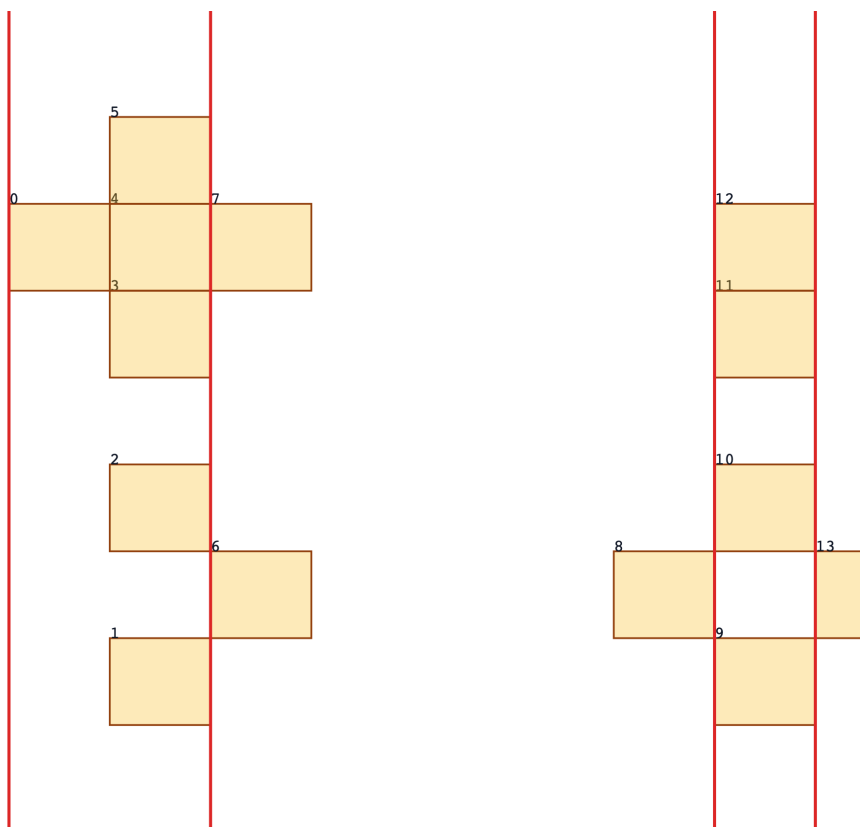


Figure 13: Best audited unit-square stabbing construction from the final matrix.

optimum. Coordinate mutation can still help when it crosses a critical endpoint and therefore changes several incidence relations at once.

7.4 Why improvement stopped

The unit-square plateau is strongly discrete. A local move usually preserves the same critical-line incidence matrix and therefore leaves both τ and τ^* unchanged. A threshold-crossing move often changes the integer and fractional optima together. The search operates on an integer coordinate grid with 8–24 accepted squares, so it explores only a small part of the realizable incidence landscape. Moreover, the generic model learns a serialization of coordinates; it has no explicit equivariance under translation, relabeling, or affine normalization.

The observed $3/2$ -type behavior is compact and repeatedly rediscovered, but the code has no neutral, general composition rule that combines such blocks without allowing one line to stab multiple blocks. That cross-block line is the geometric reason direct sums fail to amplify the set-cover gap.

Table 9: Descriptive component marginals for Guillotine. Each level pools nine distinct component cells.

Factor	Level	Mean	Median	Min.	Max.	IQR
Representation	<code>recursive_obstruction_grammar</code>	0.2556	0.2500	0.2500	0.2727	[0.2500, 0.2609]
	<code>sequence_pair_packing</code>	0.2389	0.2500	0.1538	0.3000	[0.1765, 0.2727]
	<code>rect_direct_disjoint</code>	0.2383	0.2500	0.1667	0.3333	[0.1818, 0.2727]
Local Search	<code>witness_breaking</code>	0.2810	0.2727	0.2500	0.3333	[0.2667, 0.3000]
	<code>packing_resize</code>	0.2534	0.2500	0.2353	0.2727	[0.2500, 0.2500]
	<code>recursive_gadget_assembly</code>	0.1983	0.1818	0.1538	0.2500	[0.1739, 0.2500]
Surrogate	<code>depth_limited_dp</code>	0.2560	0.2667	0.1765	0.3333	[0.2500, 0.2727]
	<code>first_cut_obstruction</code>	0.2423	0.2500	0.1538	0.3000	[0.2500, 0.2727]
	<code>k_subset_nonseparability</code>	0.2344	0.2500	0.1667	0.2727	[0.2353, 0.2500]

8 Problem 3: recursive guillotine hardness

8.1 Objective and exact verifier

For a pairwise interior-disjoint rectangle family $\mathcal{R}$,

$$\text{score}_{\text{guill}}(\mathcal{R}) = 1 - \frac{\text{save}(\mathcal{R})}{|\mathcal{R}|},$$

where $\text{save}(\mathcal{R})$ is the maximum number of rectangles preserved by an optimal recursive sequence of horizontal and vertical guillotine cuts. The exact dynamic program enumerates coordinate-induced cuts on rectangle-subset states. Its certificate stores the exact saved and destroyed counts, the optimal recursive cut tree, first-cut diagnostics, threshold diagnostics, state count, and solver status.

8.2 All 27 trajectories

The guillotine matrix shows that witness use matters more than a more elaborate generator label. `witness_breaking` averages 0.2810, compared with 0.2534 for `packing_resize` and only 0.1983 for `recursive_gadget_assembly`. The latter often adds or copies local blocks without closing the recursive escape route and is actively harmful in this matrix. Among surrogates, `depth_limited_dp` is strongest on average (0.2560), while `k_subset_nonseparability` is weakest (0.2344) despite being more elaborate. The recursive-obstruction representation is stable but capped between 0.25 and 0.2727; direct disjoint search is more variable and contains both weaker rows and the unique 1/3 record. This is a useful distinction between robust initialization and record potential.

8.3 What worked

Guillotine search improves when the representation supplies non-sliceable local structure and the mutation operator responds to actual separable subsets. Witness breaking and packing resize can close

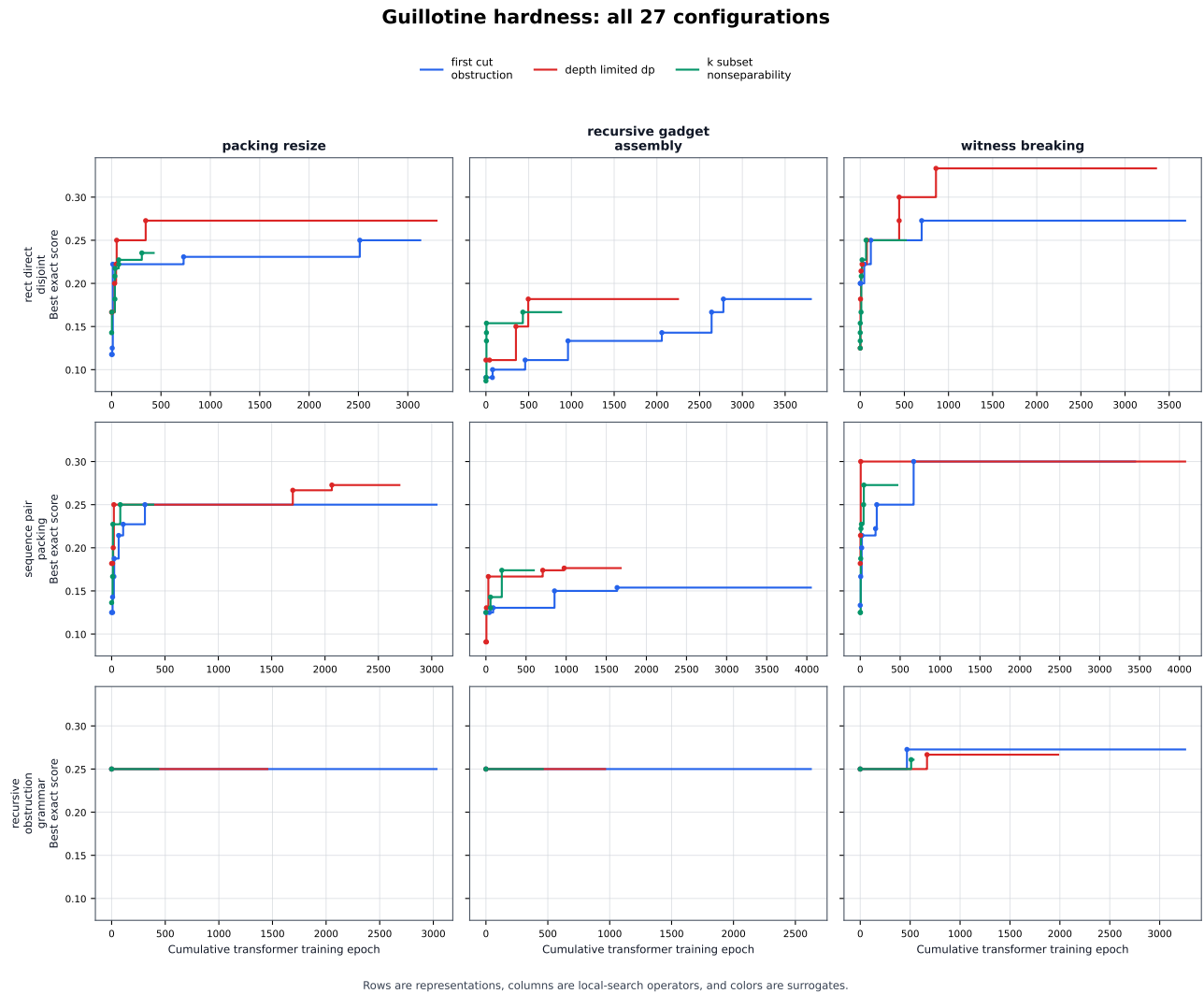


Figure 14: Guillotine trajectories against cumulative transformer-training epoch.

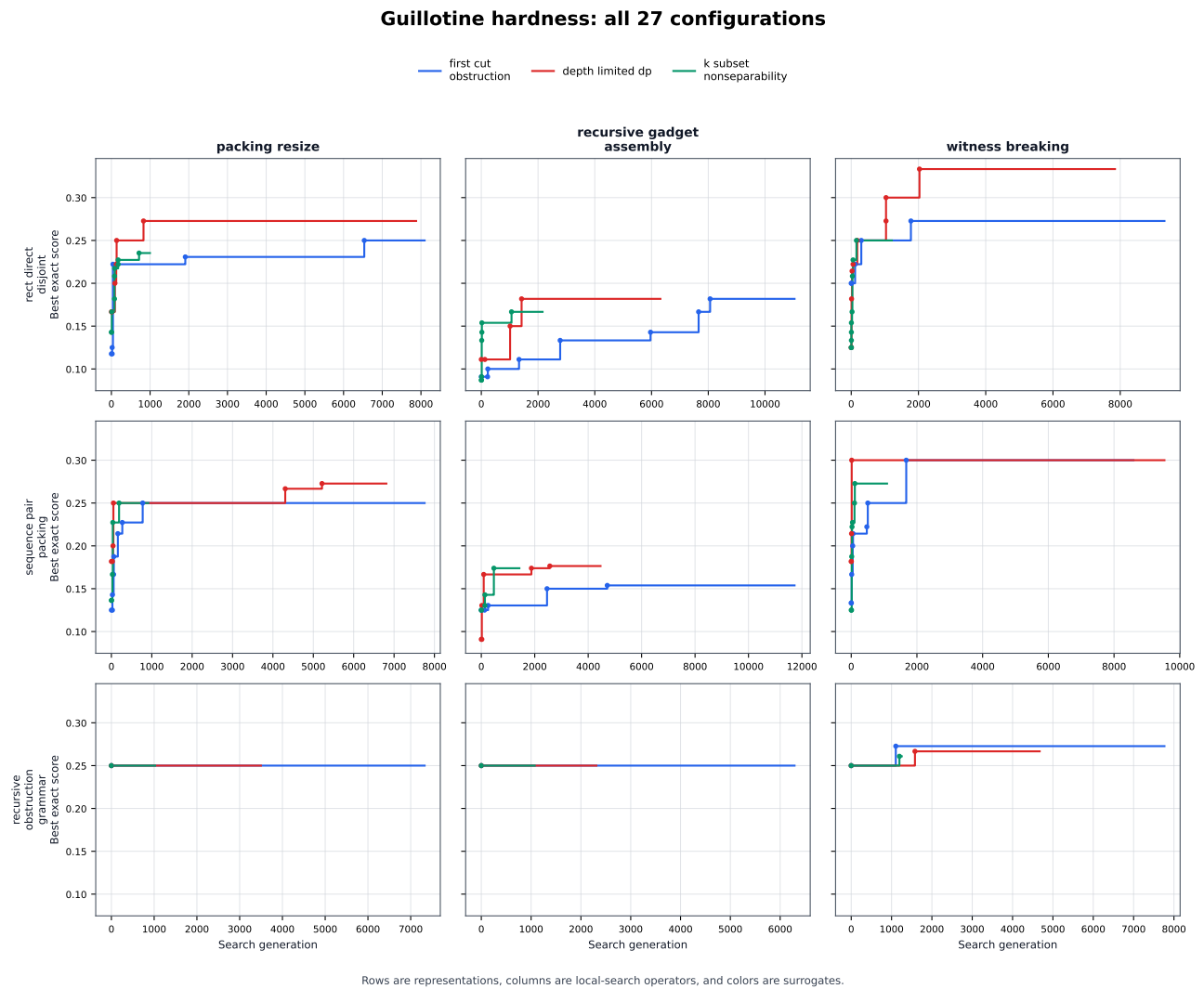


Figure 15: Guillotine trajectories against search generation.

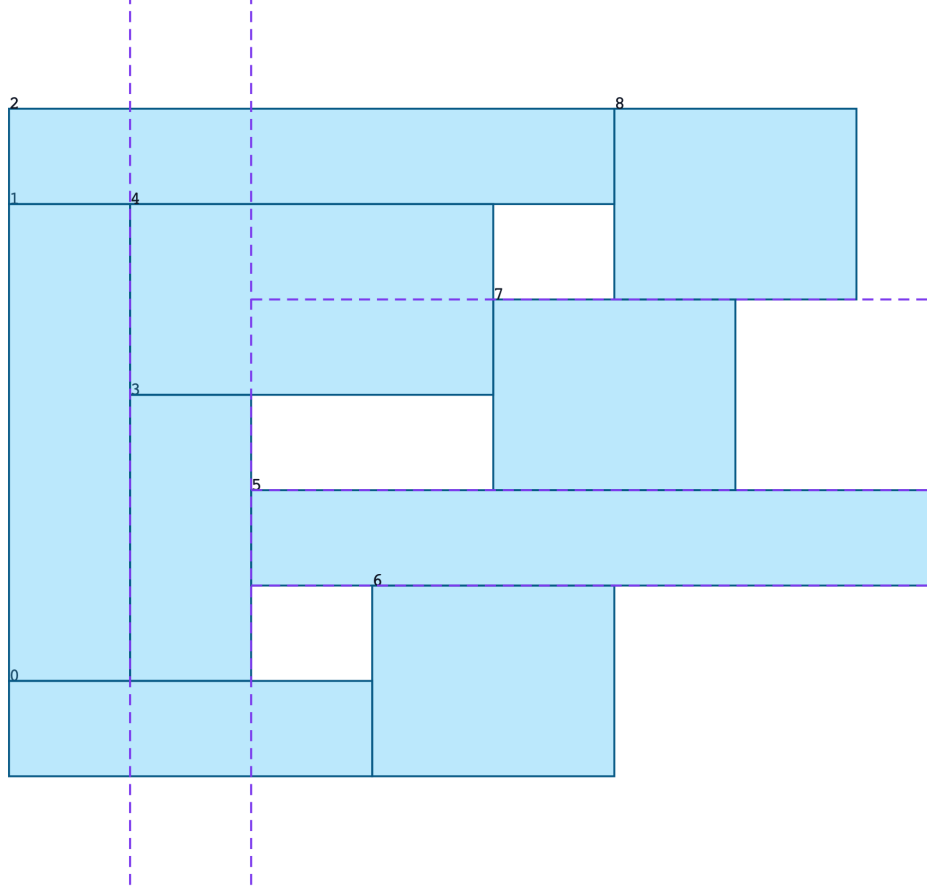


Figure 16: Best audited guillotine construction from the final matrix. The exact cut tree in the certificate, not the drawing, establishes the saved and destroyed counts.

easy projection gaps; the depth-limited dynamic program is more aligned with the exact recursive objective than first-cut obstruction. The k -subset surrogate adds useful large-subset pressure, although it is sampled above 16 rectangles.

8.4 Why improvement stopped

Guillotine hardness is the clearest case of surrogate mismatch.

1. **First-cut obstruction is not recursive obstruction.** A layout can force an expensive first cut while leaving two easy subinstances.
2. **Depth four is still shallow.** The depth-limited surrogate returns all remaining rectangles as saveable at its horizon, favoring layouts whose weakness appears just beyond that depth.
3. **Large- n threshold tests are sampled.** The deterministic subset schedule is reproducible, but 1024 subsets cannot represent all $\binom{n}{\lfloor n/2 \rfloor + 1}$ possibilities. High sampled nonseparability can coexist

with a large exact saved set.

4. **The score has coarse rational steps.** With n rectangles the score changes in increments of $1/n$. Mutations that add easy filler rectangles can make the layout look larger while reducing the destroyed fraction.
5. **Exact feedback arrives only on elites.** The exact cut tree is saved in a certificate but is not fully converted into a differentiable or dense training signal for every candidate.

The fixed spiral topology used by one representation is also a methodological limitation: it can improve the chance of seeing a local obstruction, but it narrows the search toward one family. The representation comparison must therefore be read as “which supplied search family worked best,” not as a neutral estimate over all disjoint rectangle packings.

9 Cross-problem explanation of the barriers

9.1 Prioritized diagnosis

Table 10 ranks the limitations by their likely leverage on future search. The ranking is a mechanistic interpretation of the completed matrix, not a causal effect estimate: there is one stochastic run per cell.

9.2 More epochs cannot repair a closed search space

The transformer samples only instances expressible by the current tokenizer and repaired by the current representation. If no sequence of valid local or model-generated moves reaches a stronger structural family, additional training concentrates probability inside the same basin. A flat learning curve after many successful training events is evidence of this closure. Moreover, the implementation repeatedly trains fresh short-lived models rather than continuing one model, so cumulative epochs do not accumulate weights or optimizer state across events.

9.3 The exact objectives are discontinuous

All three objectives are ratios of optimization values. Small geometric changes often do nothing; the next change may alter both numerator and denominator. Gradient-like local pressure is therefore weak. This also explains why a surrogate can improve smoothly while the exact record remains flat for thousands of generations.

9.4 Repair is both necessary and destructive

Repair enforces triangle freedom, congruent-square validity, disjointness, size bounds, and canonical order. It prevents invalid training data but can erase the coordinated multi-object change proposed by the model. The event logs record repairs and invalid samples; high repair rates indicate that the model is learning a syntax larger than the feasible search manifold.

9.5 The learned representation is mostly syntactic

The transformer is trained on serialized coordinates and permutation-like descriptions rather than on canonical conflict graphs, critical-line incidence structures, LP dual supports, or exact cut trees. Canonicalization removes some trivial equivalences, but many geometrically or combinatorially equivalent candidates still have different strings. This weakens sample efficiency and helps explain why repeated training can produce syntactically valid but structurally redundant candidates.

9.6 Budget enforcement is too coarse

The wall-clock budget is checked between search generations, not inside every surrogate or exact-solver call. A single difficult maximum-independent-set, set-cover, or guillotine-DP evaluation can therefore run beyond the nominal budget and approach the Slurm limit before the row writes its summary. This creates both a robustness risk and a small fairness distortion among cells. Production runs should enforce per-candidate solver deadlines, flush events periodically, and reserve an explicit finalization window.

Table 10: Prioritized limitations supported by the 81-run evidence. “Likely leverage” estimates how strongly fixing the limitation could change the reachable constructions, not how easy the fix would be.

Limitation	Evidence and consequence	Likely leverage
Closed representation families	Entire component families collapse to one score—for example, all nine quadratic-program MISR cells finish at 4/3, and six recursive-grammar guillotine cells remain at 1/4. Search cannot discover a structure its representation and repair operators cannot express.	Very high
Weak exact-witness feedback	Surrogates rank most candidates, while exact primal, dual, stabbing, and cut-tree witnesses are not converted into dense move-level feedback. Learned-model samples account for only 0, 2, and 6 strict records in MISR, unit-square, and guillotine.	Very high
Archive collapse and model restart	Median duplicate counts reach 17.16, 9.46, and 7.25 per generation. Each training event fits a fresh three-epoch model, so optimizer state and learned features do not accumulate across events.	Very high
No composition mechanism	The best certificates have only 12, 14, and 9 objects. The code searches finite witnesses but has no reliable operator that combines two certified blocks while preserving the relevant ratio. A record therefore does not automatically generate a stronger family.	Very high
Discrete exact objectives	Small geometric changes usually leave the exact ratio unchanged, while the next useful change may require several coordinated objects. This creates sparse rewards and explains the 74–98% median post-best plateaus.	High
Proposal–repair mismatch	Repair guarantees valid triangle-free, unit-square, or disjoint-rectangle instances, but it can undo a coordinated proposal. A high valid-sample rate can therefore coexist with little exact progress.	High
Coordinate serialization rather than invariants	The learned model mainly sees serialized coordinates or permutations. Graph isomorphism, endpoint order, LP-dual support, and cut-tree structure are only partially exposed, so capacity is spent relearning equivalent encodings.	Medium–high
Finite size, grid, and verifier regimes	Object-count and coordinate ranges are bounded to keep exact verification tractable. More epochs at the same scale do not test whether a qualitatively different obstruction appears only at larger n or finer order type.	Medium
Unreplicated cells and unequal solver cost	One run per cell prevents error bars over stochastic repetitions. Solver-dependent throughput and generation-level budget checks also give cells unequal numbers of proposals under the same nominal wall time.	Methodological
Incomplete source provenance	The event label <code>model.sample</code> does not separate a direct transformer sample from the character n -gram backfill. Source-attribution plots are therefore conservative learned-path diagnostics, not a clean transformer ablation.	Measurement

9.7 Archive collapse wastes late training

The training corpus is drawn from the top surrogate-ranked archive without a quality-diversity constraint. Once the archive concentrates around one basin, repeated fresh-model fits generate many duplicates or repaired variants of existing candidates. Duplicate count and valid model-sample yield should therefore be treated as primary learning diagnostics, not merely bookkeeping.

9.8 Finite-scale search may hide a different regime

Exact verification constrains the practical object counts, coordinate grids, and subset limits used during search. Those bounds are scientifically necessary for trustworthy certificates, but they also mean that all 24-hour cells explore a finite regime. A larger obstruction that requires a new order type, denominator, or recursive scale may be absent from every current representation. Progressive size curricula and cross-size transfer should therefore be tested separately from simply extending a fixed-size run.

9.9 One random run per cell limits causal claims

The full factorial permits complete coverage of component combinations, but without repeated seeds an observed difference between two cells contains both a component effect and run-specific stochasticity. Main-effect means pool nine distinct cells and are useful for prioritization, yet they are not standard errors. This is the principal statistical limitation of the final matrix.

9.10 What better sampling would actually change

Increasing the number of samples from the same archive-conditioned generator would mostly increase duplicates and repaired variants. A useful sampling upgrade would instead combine:

1. a quality-diversity archive indexed by problem-specific structural features rather than surrogate score alone;
2. novelty measured after canonicalization, so translated or relabeled copies do not consume the sampling budget;
3. uncertainty- or disagreement-based exact-audit selection, allocating expensive verifier calls to candidates for which the surrogate is least trusted;
4. adaptive mixtures of model samples, local mutations, and fresh random restarts based on their recent rates of valid and record-improving output; and
5. stratified sampling across object count, denominator, and obstruction type to avoid collapse around one easy rational score.

9.11 What more complex local search should mean

Adding a longer list of arbitrary mutations is unlikely to help. The useful complexity is coordinated and verifier-driven: multi-object moves that attack an LP dual bottleneck, a stabbing line, a separable subset, or a branch in the exact guillotine cut tree. Beam or population search should retain temporary score decreases when they increase structural novelty or close an exact escape witness. This is qualitatively different from applying more single coordinate perturbations.

9.12 The experiment discovers witnesses, not general theorems

An exact certificate proves the objective value of one finite construction. It does not prove an asymptotic family, optimality over all constructions, or a universal lower bound. The missing scientific step for all three problems is composition: a rule that scales a certified finite obstruction while preserving or improving the relevant ratio.

10 Recommendations arising from the data

These recommendations describe what a successor project should change; they are not additional claims from the completed run.

10.1 MISR

1. Learn and mutate canonical conflict graphs jointly with geometric realizations, so graph-equivalent coordinate encodings do not dominate the archive.
2. Use exact LP dual support and the exact independent set to propose coordinated multi-rectangle moves.
3. Test composition operators explicitly by measuring incremental changes in both α^* and α .
4. Separate unrestricted and triangle-free record tracks; their structural barriers differ.

10.2 Unit-square stabbing

1. Search canonical critical-line incidence matrices first, then solve a geometric realization problem.
2. Add incidence-preserving block moves and cross-block separation rules to prevent one line from collapsing a composed construction.
3. Increase coordinate resolution only together with canonicalization; otherwise the model spends capacity on translations and equivalent endpoint orders.
4. Retain exact primal and dual witnesses in the learning features.

10.3 Guillotine

1. Convert the exact optimal cut tree and saved set into explicit mutation targets after every record improvement.
2. Replace fixed-depth scoring with adaptive exact scoring on medium-sized elites and a learned estimator of exact saved fraction for larger candidates.
3. Diversify obstruction generators beyond one fixed topology and report generator family as an explicit experimental factor.
4. Penalize easy filler rectangles and optimize destroyed count and family size lexicographically before taking the ratio.

11 Reproducibility and artifact map

Raw event streams are retained on the HPC run root rather than committed to Git. On that run root, the compact analysis is regenerated with:

```
python3 scripts/compact_main_matrix_events.py \  
  runs/main_81_24h_20260709_1623/main_results \  
  --out-dir runs/main_81_24h_20260709_1623/postprocess/analysis \  
  --expected-rows 81 --default-model-epochs 3 --strict
```

Plots and descriptive tables are regenerated with:

```
python3 scripts/plot_main_matrix_analysis.py \  
  docs/assets/main_81_24h_20260709_1623/postprocess \  
  --out-dir docs/assets/main_81_24h_20260709_1623/analysis
```

Generated TeX tables are rebuilt from the committed compact evidence with:

```
python3 scripts/generate_main_matrix_tex.py \  
  docs/assets/main_81_24h_20260709_1623/postprocess \  
  --plot-table-dir docs/assets/main_81_24h_20260709_1623/analysis \  
  --audit-csv docs/assets/main_81_24h_20260709_1623/postprocess/audit.csv \  
  --out-dir docs/assets/main_81_24h_20260709_1623/report
```

The analysis manifest records hashes for the compaction script and every compact CSV. The repository snapshot contains the compact trajectory data, the 81-row metrics, audit table, best certificates and renderings, generated figures, and this report. Re-running the exact audit requires the HPC run root because the original summaries retain their certificate paths. The raw event streams remain the source of truth for candidate-level reconstruction.

A Complete 81-cell table

Table 11: Complete 81-cell result table. Identifiers are shortened here; the full names appear in the design and marginal-effect tables. Plateau is the fraction of completed search generations after the final strict score improvement.

Prob.	Rep.	Local	Surrogate	Best	Gain	Epoch	h	Plat. (%)	n
MISR	endpoint_seq	lp.dual	exact_lp_gap	1.2500	0.2500	258	0.32	97.4	8
MISR	endpoint_seq	lp.dual	conflict_ proxy	1.2500	0.2500	120	12.19	97.9	23
MISR	endpoint_seq	lp.dual	tf_exact_gap	1.2500	0.2500	111	0.13	98.9	8
MISR	endpoint_seq	coeff_pivot	exact_lp_gap	1.3333	0.3333	3141	7.05	49.2	15
MISR	endpoint_seq	coeff_pivot	conflict_ proxy	1.2500	0.2500	12	0.02	99.6	19
MISR	endpoint_seq	coeff_pivot	tf_exact_gap	1.2500	0.2500	159	0.15	97.7	24
MISR	endpoint_seq	seq_pivot	exact_lp_gap	1.2500	0.2500	213	0.65	96.2	8
MISR	endpoint_seq	seq_pivot	conflict_ proxy	1.2500	0.2500	27	0.04	99.5	9
MISR	endpoint_seq	seq_pivot	tf_exact_gap	1.3333	0.3333	4989	14.97	26.6	9
MISR	qp_rect	lp.dual	exact_lp_gap	1.3333	0.2083	567	0.96	92.1	8
MISR	qp_rect	lp.dual	conflict_ proxy	1.3333	0.3333	87	0.15	98.7	8
MISR	qp_rect	lp.dual	tf_exact_gap	1.3333	0.3333	111	0.19	98.4	8
MISR	qp_rect	coeff_pivot	exact_lp_gap	1.3333	0.3333	75	0.29	98.6	8
MISR	qp_rect	coeff_pivot	conflict_ proxy	1.3333	0.3333	192	0.33	97.2	8
MISR	qp_rect	coeff_pivot	tf_exact_gap	1.3333	0.3333	1104	2.11	84.1	8
MISR	qp_rect	seq_pivot	exact_lp_gap	1.3333	0.3333	510	1.19	92.2	8
MISR	qp_rect	seq_pivot	conflict_ proxy	1.3333	0.3333	1494	3.02	79.2	8
MISR	qp_rect	seq_pivot	tf_exact_gap	1.3333	0.3333	336	0.60	95.2	8
MISR	triangle_ free	lp.dual	exact_lp_gap	1.3750	0.3750	81	0.08	98.7	11
MISR	triangle_ free	lp.dual	conflict_ proxy	1.3333	0.3333	0	0.00	100.0	8
MISR	triangle_ free	lp.dual	tf_exact_gap	1.3750	0.2500	24	0.03	99.6	11
MISR	triangle_ free	coeff_pivot	exact_lp_gap	1.4000	0.4000	4782	14.48	25.1	14
MISR	triangle_ free	coeff_pivot	conflict_ proxy	1.3333	0.3333	51	0.05	99.3	8
MISR	triangle_ free	coeff_pivot	tf_exact_gap	1.3750	0.3750	156	0.15	97.5	11
MISR	triangle_ free	seq_pivot	exact_lp_gap	1.5000	0.3750	1860	3.12	70.7	12
MISR	triangle_ free	seq_pivot	conflict_ proxy	1.1250	0.1250	66	0.06	99.0	9
MISR	triangle_ free	seq_pivot	tf_exact_gap	1.3750	0.3750	231	0.22	96.4	11
Unit sq.	line_sq_inc	coord_mut	exact_stab_ gap	1.5000	0.5000	3915	11.18	40.2	21
Unit sq.	line_sq_inc	coord_mut	greedy_lp	1.2500	0.1071	4587	16.21	21.8	24
Unit sq.	line_sq_inc	coord_mut	incidence_ stats	1.3333	0.3333	873	1.01	87.3	9
Unit sq.	line_sq_inc	primal_dual	exact_stab_ gap	1.4286	0.4286	276	0.57	95.4	24
Unit sq.	line_sq_inc	primal_dual	greedy_lp	1.2069	0.2069	1458	3.35	74.4	24
Unit sq.	line_sq_inc	primal_dual	incidence_ stats	1.2500	0.2500	855	1.42	86.4	13

Prob.	Rep.	Local	Surrogate	Best	Gain	Epoch	h	Plat. (%)	n
Unit sq.	line_sq_inc	sqstab_hill	exact_stab_gap	1.5000	0.5000	3003	9.29	48.9	21
Unit sq.	line_sq_inc	sqstab_hill	greedy_lp	1.3636	0.2208	2103	6.05	62.4	24
Unit sq.	line_sq_inc	sqstab_hill	incidence_stats	1.3333	0.3333	2781	7.03	54.7	15
Unit sq.	sqstab_grid	coord_mut	exact_stab_gap	1.5000	0.3571	117	0.35	98.5	14
Unit sq.	sqstab_grid	coord_mut	greedy_lp	1.3636	0.2208	6777	23.68	0.8	20
Unit sq.	sqstab_grid	coord_mut	incidence_stats	1.3158	0.1158	147	0.14	98.0	10
Unit sq.	sqstab_grid	primal_dual	exact_stab_gap	1.4286	0.2857	171	9.83	97.4	22
Unit sq.	sqstab_grid	primal_dual	greedy_lp	1.3636	0.3636	732	1.02	88.9	16
Unit sq.	sqstab_grid	primal_dual	incidence_stats	1.3333	0.3333	1839	2.76	74.3	10
Unit sq.	sqstab_grid	sqstab_hill	exact_stab_gap	1.5000	0.3571	375	0.46	94.8	19
Unit sq.	sqstab_grid	sqstab_hill	greedy_lp	1.3462	0.3462	3087	6.68	55.5	22
Unit sq.	sqstab_grid	sqstab_hill	incidence_stats	1.3333	0.3333	0	0.00	100.0	9
Unit sq.	square_direct	coord_mut	exact_stab_gap	1.4286	0.4286	4284	11.83	33.9	23
Unit sq.	square_direct	coord_mut	greedy_lp	1.2500	0.2500	2181	3.75	65.0	23
Unit sq.	square_direct	coord_mut	incidence_stats	1.2500	0.2500	48	0.04	99.3	9
Unit sq.	square_direct	primal_dual	exact_stab_gap	1.4286	0.2857	405	0.39	94.1	19
Unit sq.	square_direct	primal_dual	greedy_lp	1.2000	0.2000	2931	5.79	55.2	24
Unit sq.	square_direct	primal_dual	incidence_stats	1.2500	0.2500	1287	1.17	82.6	13
Unit sq.	square_direct	sqstab_hill	exact_stab_gap	1.4516	0.3088	2124	3.68	68.3	24
Unit sq.	square_direct	sqstab_hill	greedy_lp	1.4286	0.2857	276	0.28	95.7	22
Unit sq.	square_direct	sqstab_hill	incidence_stats	1.3158	0.3158	4845	11.90	33.1	13
Guill.	rect_direct	pack_resize	depth_dp	0.2727	0.1061	345	1.92	89.5	11
Guill.	rect_direct	pack_resize	first_cut	0.2500	0.1324	2514	17.91	19.3	8
Guill.	rect_direct	pack_resize	k_subset	0.2353	0.0924	306	16.72	28.3	17
Guill.	rect_direct	gadget_assembly	depth_dp	0.1818	0.0707	495	4.10	77.5	22
Guill.	rect_direct	gadget_assembly	first_cut	0.1818	0.0909	2778	13.83	27.0	11
Guill.	rect_direct	gadget_assembly	k_subset	0.1667	0.0797	432	11.02	50.7	18
Guill.	rect_direct	witness_break	depth_dp	0.3333	0.2083	858	5.81	74.1	9
Guill.	rect_direct	witness_break	first_cut	0.2727	0.0727	696	3.06	80.9	11
Guill.	rect_direct	witness_break	k_subset	0.2500	0.1250	66	2.74	87.2	8
Guill.	obstruction_grammar	pack_resize	depth_dp	0.2500	0.0000	0	0.00	100.0	8
Guill.	obstruction_grammar	pack_resize	first_cut	0.2500	0.0000	0	0.00	100.0	8
Guill.	obstruction_grammar	pack_resize	k_subset	0.2500	0.0000	0	0.00	100.0	8

Prob.	Rep.	Local	Surrogate	Best	Gain	Epoch	h	Plat. (%)	n
Guill.	obstruction_grammar	gadget_assembly	depth_dp	0.2500	0.0000	0	0.00	100.0	8
Guill.	obstruction_grammar	gadget_assembly	first_cut	0.2500	0.0000	0	0.00	100.0	8
Guill.	obstruction_grammar	gadget_assembly	k_subset	0.2500	0.0000	0	0.00	100.0	8
Guill.	obstruction_grammar	witness_break	depth_dp	0.2667	0.0167	669	8.10	66.2	15
Guill.	obstruction_grammar	witness_break	first_cut	0.2727	0.0227	468	2.54	85.8	11
Guill.	obstruction_grammar	witness_break	k_subset	0.2609	0.0109	510	22.87	4.6	23
Guill.	seq_pair_pack	pack_resize	depth_dp	0.2727	0.0909	2064	17.56	23.4	11
Guill.	seq_pair_pack	pack_resize	first_cut	0.2500	0.1250	312	1.76	90.0	8
Guill.	seq_pair_pack	pack_resize	k_subset	0.2500	0.1136	81	5.22	79.3	24
Guill.	seq_pair_pack	gadget_assembly	depth_dp	0.1765	0.0856	975	15.52	42.5	17
Guill.	seq_pair_pack	gadget_assembly	first_cut	0.1538	0.0288	1635	5.29	59.8	13
Guill.	seq_pair_pack	gadget_assembly	k_subset	0.1739	0.0489	198	8.97	67.0	23
Guill.	seq_pair_pack	witness_break	depth_dp	0.3000	0.1182	6	0.05	99.8	10
Guill.	seq_pair_pack	witness_break	first_cut	0.3000	0.1667	669	3.51	80.5	10
Guill.	seq_pair_pack	witness_break	k_subset	0.2727	0.1477	45	2.12	89.9	22

B Interpretation guide

Table 12: How to interpret the main analysis fields.

Field	Meaning
Best exact score	Highest score recomputed by the problem’s exact scorer and saved in the row checkpoint/summary.
Initial best	First nontrivial exact score visible in the event stream.
Score gain	Final best minus initial best; it measures within-run discovery, not comparison with historical experiments.
Best model epoch	Cumulative successful transformer-training epochs before the event that first exposed the final best.
Time to best	Wall-clock seconds from row start to the first saved final-best certificate.
Plateau fraction	Fraction of completed search generations occurring after the last strict best-score improvement.
Model improvement	Strict improvement whose audited candidate had <code>source_type=model_sample</code> .
Fallback	Emergency randomized export attempted only when a guillotine row would otherwise end without any certificate; unused in this run.

C Evidence boundary

The report excludes epsilon-net and graph-separation experiments, controls from earlier arrays, all square-stabbing-14-9 package evidence, historical scores without certificates, and warm-start runs. Those artifacts may be discussed elsewhere, but they are not mixed into this 81-cell final matrix.